

Cancer — Types, Prevention, and Treatment

1. Introduction and Cancer Biology

Cancer is a group of diseases characterised by uncontrolled growth and spread of abnormal cells. If the spread is not controlled, it can result in death. Cancer is caused by changes (mutations) in the DNA within cells that alter normal cellular regulation of growth, division, differentiation, and death.

Hallmarks of cancer (Hanahan and Weinberg): sustaining proliferative signalling, evading growth suppressors, resisting cell death (apoptosis), enabling replicative immortality, inducing angiogenesis, activating invasion and metastasis, reprogramming energy metabolism, and evading immune destruction.

Global burden: approximately 19.3 million new cases and 10 million cancer deaths occurred in 2020. Lung, breast, colorectal, prostate, and cervical cancers are among the most common. In India, approximately 1.46 million new cancer cases were diagnosed in 2022, with cervical, breast, oral cavity, lung, and colorectal cancers being most prevalent.

2. Cancer Epidemiology in India

India's cancer profile differs from Western countries. Tobacco-related cancers (oral cavity, larynx, oesophagus, lung) account for approximately 27% of cases in men. Cervical cancer (HPV-related) and breast cancer are the leading cancers in women.

Oral cancers: India has one of the highest incidences of oral cancer globally due to the widespread use of smokeless tobacco (gutka, khaini, pan masala, betel quid with tobacco). Oral cancer accounts for approximately 30% of all cancers in India.

Regional variation: North-Eastern India has high rates of oesophageal cancer (Assam, Nagaland). Punjab has high cervical cancer rates. Stomach cancer is more prevalent in South India (Kerala, Tamil Nadu). Gallbladder cancer is particularly common in the Gangetic belt (Bihar, Uttar Pradesh, Delhi — related to gallstones and possibly contaminated water).

Infrastructure: India has approximately 450 cancer hospitals and several dedicated Regional Cancer Centres (RCCs) in each state. Tata Memorial Centre (Mumbai), AIIMS (Delhi), PGIMER (Chandigarh), and Kidwai Memorial (Bangalore) are national reference centres.

3. Common Cancers — Breast Cancer

Breast cancer is the most common cancer in women globally and in India, accounting for 28% of all cancers in Indian women. Incidence is rising, particularly in urban India.

Risk factors: female sex, age, BRCA1/BRCA2 mutations (high penetrance — 50-85% lifetime risk), family history, early menarche, late menopause, nulliparity, HRT, obesity, alcohol, radiation exposure, dense breast tissue.

Symptoms: breast lump (most common — painless, hard, irregular borders), nipple discharge (bloody or clear), skin changes (dimpling — peau d'orange, erythema), nipple inversion, axillary lymphadenopathy.

Staging: TNM staging (I-IV). Molecular subtypes: Luminal A (ER+, HER2-, low grade — best prognosis), Luminal B (ER+, HER2± — intermediate), HER2-enriched (HER2+, ER- — treated with trastuzumab), Triple Negative (TNBC — ER-, PR-, HER2- — worst prognosis, chemotherapy backbone).

Treatment: Surgery (breast-conserving surgery + radiotherapy = mastectomy in terms of survival for eligible patients. Sentinel lymph node biopsy reduces morbidity). Systemic: chemotherapy, endocrine therapy (tamoxifen, aromatase inhibitors), anti-HER2 therapy (trastuzumab, pertuzumab, T-DM1, trastuzumab-deruxtecan), CDK4/6 inhibitors, PARP inhibitors, immunotherapy (pembrolizumab for TNBC).

4. Cervical Cancer

Cervical cancer is the second most common cancer in Indian women and a major cause of cancer death in low- and middle-income countries. Nearly all cases are caused by persistent infection with high-risk HPV (mainly types 16 and 18).

Risk factors: early sexual debut, multiple sexual partners, high parity, long-term OCP use, smoking, HIV infection, not being vaccinated against HPV. Most HPV infections resolve spontaneously but persistent infection over years can cause cervical intraepithelial neoplasia

(CIN) and ultimately cancer.

Screening: Pap smear (cytology) every 3 years from 21-65; HPV DNA testing (more sensitive) every 5 years from 30-65. Visual Inspection with Acetic acid (VIA) — low-cost alternative in low-resource settings. India's National Cervical Cancer Screening Programme recommends VIA for women 30-59.

HPV vaccination: Gardasil 9 (bivalent/quadrivalent/9-valent). India launched school-based HPV vaccination for girls aged 9-14 in 2023. Highly cost-effective; prevents ~70-90% of cervical cancers.

Treatment: early (IA-IIA) — surgery (radical hysterectomy) or radiotherapy. Locally advanced (IIB-IVA): concurrent chemoradiotherapy (CCRT) with weekly cisplatin. Metastatic: bevacizumab + chemotherapy (KEYNOTE-826 — pembrolizumab + platinum-based chemo + bevacizumab improves survival in PD-L1 positive).

5. Lung Cancer

Lung cancer is the leading cause of cancer death globally. Two main types: Non-small cell lung cancer (NSCLC, 85%) — adenocarcinoma (most common, peripheral, increasingly in non-smokers, harbouring targetable mutations), squamous cell carcinoma (central, smoking-related), large cell carcinoma. Small cell lung cancer (SCLC, 15%) — neuroendocrine, extremely aggressive, early metastasis.

Risk factors: tobacco smoking (most important — 85% of cases; includes hookah, bidi — both common in India), occupational exposures (asbestos, radon, polycyclic aromatic hydrocarbons), indoor air pollution, genetic susceptibility.

Molecular drivers in NSCLC: EGFR mutations (10-15% in Western, 40-50% in East Asian patients — treated with EGFR-TKIs: osimertinib, erlotinib), ALK rearrangements (3-5% — alectinib, crizotinib), ROS1, MET, BRAF, RET, KRAS G12C (sotorasib). PD-L1 expression guides immunotherapy use (pembrolizumab).

Treatment: early NSCLC (Stage I-II) — surgical resection (lobectomy preferred) ± adjuvant chemotherapy/osimertinib (EGFR+). Locally advanced (Stage III) — concurrent chemoradiotherapy ± durvalumab maintenance. Metastatic — molecular-targeted therapy if driver mutation present, otherwise platinum-based chemotherapy ± immunotherapy.

6. Gastrointestinal Cancers

Colorectal cancer (CRC): third most common cancer globally. Risk factors: diet (red meat, processed meat, low fibre), sedentary lifestyle, obesity, diabetes, IBD, family history, Lynch syndrome. Screening: colonoscopy every 10 years from age 45 (or faecal immunochemical test FIT annually). Treatment: surgery for localised disease; FOLFOX/CAPOX chemotherapy $\pm$ bevacizumab/cetuximab for advanced.

Stomach (Gastric) cancer: declining in most regions due to *H. pylori* eradication and food refrigeration, but still significant in South India. *H. pylori* infection is by far the most important modifiable risk factor. Test-and-treat *H. pylori* in high-risk individuals. Symptoms often late: dyspepsia, weight loss, early satiety. Signet ring cell carcinoma is particularly aggressive.

Pancreatic cancer: 5-year survival $\sim$ 12% — one of the worst prognoses of any cancer. Usually presents late (painless jaundice — head of pancreas, back pain — body/tail). Surgery (Whipple's procedure for head lesions) offers only chance of cure. Gemcitabine/nab-paclitaxel or FOLFIRINOX for advanced disease. BRCA mutations — consider olaparib.

7. Leukaemia and Lymphoma

Acute leukaemias: Acute Lymphoblastic Leukaemia (ALL) — most common childhood cancer; BFM protocol with L-asparaginase gives 85-90% cure rate in children. Acute Myeloid Leukaemia (AML) — primarily adult; 7+3 induction chemotherapy; HSCT for high-risk.

Chronic leukaemias: CML — Ph⁺ chromosome, BCR-ABL fusion protein. Imatinib (Gleevec) revolutionised CML treatment — one of oncology's greatest success stories; newer TKIs (dasatinib, nilotinib, asciminib). CLL — indolent; ibrutinib (BTK inhibitor), venetoclax (BCL-2 inhibitor) standard of care.

Lymphomas: Hodgkin Lymphoma (HL) — curable with ABVD chemotherapy (85-90% cure rate for early stages). Non-Hodgkin Lymphoma (NHL) — 60+ subtypes; DLBCL most common aggressive NHL — R-CHOP (rituximab + CHOP chemotherapy). Follicular lymphoma — indolent, relapsing, treated when symptomatic.

8. Cancer Prevention

Tobacco cessation: the single most important cancer prevention measure. Tobacco causes 30% of all cancer deaths. WHO MPOWER framework and India's COTPA legislation are key policy tools. Pharmacotherapy: varenicline (most effective), bupropion, NRT (nicotine replacement therapy).

Vaccination: HPV vaccine (cervical, oropharyngeal, anal cancer prevention); HBV vaccine (hepatocellular carcinoma prevention). Diet and lifestyle: achieve and maintain healthy weight (obesity linked to 12 cancers), Mediterranean diet, limit red/processed meat, avoid alcohol, regular physical activity (150 min/week moderate).

Sun protection: UV radiation causes >90% of skin cancers. Apply SPF 30+ broad-spectrum sunscreen, seek shade during peak UV hours, wear protective clothing and hats.

Screening programmes: breast self-examination and clinical breast examination awareness; mammography for women 40-74 every 2 years (in resource settings); Pap smear/HPV DNA for cervical cancer; VIA where colposcopy unavailable; LDCT for lung cancer screening in current/ex-heavy smokers aged 50-80.

9. Cancer Treatment Principles

Surgery: curative (resection of localised tumour with clear margins — the intention is to cure), palliative (relieve symptoms — e.g., bypass obstruction), preventive (prophylactic mastectomy in BRCA1 carriers), reconstructive, staging (sentinel node biopsy, diagnostic laparoscopy).

Radiation therapy: targeted delivery of ionising radiation to tumour. External beam RT (EBRT) — linear accelerator; IMRT/IGRT allows conformity. Stereotactic body RT (SBRT/SABR) for small lung/liver/spine lesions. Brachytherapy (internal RT) — cervical cancer, prostate. Proton therapy — reduces radiation exposure to surrounding normal tissues.

Systemic therapy: traditional cytotoxic chemotherapy (DNA damage, mitotic inhibition); targeted therapy (monoclonal antibodies, TKIs, ADCs); immunotherapy (checkpoint inhibitors: pembrolizumab, nivolumab, atezolizumab; CAR-T cells; bispecific antibodies); hormonal therapy (breast, prostate cancer); bone marrow/stem cell transplant.

10. Palliative Care and Cancer in India

Palliative care is an approach that improves the quality of life of patients and families facing life-threatening illness through the prevention and relief of suffering. It addresses physical, psychological, social, and spiritual dimensions.

India's palliative care deficit: approximately 5.5 million Indians need palliative care annually; only ~1% have access. Kerala is the model state — community-based palliative care through Neighbourhood Networks in Palliative Care (NNPC) reaches rural populations. National Palliative Care Programme guidelines were published but implementation is poor in most states.

Pain control: the WHO analgesic ladder — paracetamol/NSAIDs (Step 1) → weak opioids/tramadol (Step 2) → strong opioids/morphine (Step 3). Oral morphine is the gold standard for severe cancer pain. India's restrictive opioid regulations (NDPS Act) historically created barriers to morphine access; 2014 amendments simplified access to essential narcotic medicines.

Psychosocial support: cancer diagnosis causes significant psychological distress. Depression and anxiety affect 25-50% of cancer patients. Early referral to psycho-oncology, patient support groups, spiritual care, and social work is integral to comprehensive cancer care.

Ayushman Bharat PM-JAY: India's flagship health insurance scheme covers cancer treatment up to ₹5 lakh per family per year, including surgery, chemotherapy, and radiation therapy, at empanelled government and private hospitals. Significantly improves cancer care access for below-poverty-line families.